JOE WILSON

Generative AI Engineer | NLP| Computer Vision| Machine Learning

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PROFESSIONAL SUMMARY

Generative AI Engineer and **Computer vision** with **6.5+ years** of experience in designing enterprise AI solutions using LLMs, **RAG, Agentic AI, and Computer Vision.** Skilled in **Python, FastAPI, LangChain, OpenAI, and Claude,** with proven expertise in building scalable, production-ready AI platforms for aerospace and defense applications.

CORE COMPETENCIES

Generative AI & LLMs	GPT-4o, GPT-4, GPT-3.5 Turbo, Claude 3, Llama2, Mistral 7B, Prompt Engineering, RAG, Agentic AI, Function Calling, Instruction Fine-Tuning, LoRA, QLoRA, PEPT, RLHF, DPO, Chain-of-Thought Prompting, Few-Shot Learning, Structured Output Generation, Multimodal AI
LLM Frameworks & Orchestration	LangChain, LangGraph, LlamaIndex, Haystack, Hugging Face Transformers, OpenAI SDK, Anthropic SDK
AI Agents & Automation	CrewAI, LangGraph Agents, MCP (Model Context Protocol), Tool Calling, Multi-Agent Workflows
Vector Databases & Semantic Search	Pinecone, Weaviate, ChromaDB, FAISS, Qdrant, Sentence Transformers, OpenAI Embeddings, BM25, Dense Retrieval, ANN Search
Machine Learning & NLP	PyTorch, TensorFlow, Scikit-learn, XGBoost, Pandas, Numpy, spaCy, NLTK, OpenCV, Hugging Face Datasets
LLM Evaluation & AI Safety	Prompt Evaluation, Hallucination Detection, AI Guardrails, Responsible AI
LLMops	Weights & Biases (W&B), Docker, GitHub Actions, FastAPI, vLLM, ONNX Runtime
Cloud & AI Platforms	AWS – (EC2, ECS, EKS, S3)| Microsoft Azure – Azure AI Foundry, Azure OpenAI Service, Model Deployments.
Databases	RDBMS, MongoDB
Backend Development	Python, FastAPI, Flask, Pydantic, SQL
DevOps & Infrastructure	Docker, Kubernetes, Git, GitHub, Linux, Bash, Prometheus, Grafana
Tools & Design	Cursor, Draw.io, Eraser, UX Pilot, PuTTYgen, WinSCP

PROFESSIONAL EXPERIENCE

Generative AI Engineer

Aug 2025 – Sep 2026

Rangsons Aerospace Pvt Ltd | Bangalore, India

- Led end-to-end delivery of multiple Generative AI initiatives from ideation through Proof of Concept (PoC) to production deployment enabling enterprise adoption of AI-powered knowledge assistants and workflow automation.
- Developed rapid Generative AI prototypes using OpenAI , Anthropic Claude, RAG, and Prompt Engineering, accelerating business validation cycles and reducing prototype development time.
- Owned the complete AI application lifecycle, including solution architecture, backend API development, cloud deployment, monitoring, security, and production support.
- Architected scalable RAG pipelines using LangChain and vector databases (Pinecone, ChromaDB, pgvector, Weaviate) to improve response accuracy using enterprise knowledge sources.

Computer Vision Engineer
Jan 2024 - Jul 2025

Rangsons Aerospace Pvt Ltd | Bangalore, India

- Engineered AI-powered computer vision solutions for defence applications using advanced object detection models, contributing to mission target recognition capabilities.

Associate - Data Management
Aug 2021 - Dec 2023

AGEYE Technologies Pvt Ltd | Bangalore, India

- Designed and implemented an end-to-end image analysis pipeline integrating image processing, feature extraction, data analytics, visualization, and data management to support precision agriculture and plant growth monitoring.

Software Developer
May 2020 - Mar 2021

Thinkinfinity Technology | Chennai, India

RPA - Developer
April 2019 - Jan 2020

Omega Healthcare Management Service Pvt Ltd | Tiruchirappalli, India

KEY PROJECTS

Agentic AI Chabot with RAG, HITL & CI/CD Deployment on AWS EC2
April 2026 – Aug 2026

- Developed an agentic AI Chatbot using **Python, ReactJS, LangGraph**, and **LangChain,** integrating, LLM-powered conversations with tool calling and external API integrations.
- Implemented a **Retrieval-Augmented Generation (RAG)** workflow to retrieve relevant information from a knowledge base and provide context-grounded responses to user queries.
- Integrated **Human-in-the-loop (HITL)** workflows to pause selected agent actions for human review and approval before proceeding with sensitive or consequential operations.
- Designed agent workflows using **LangGraph**, enabling conditional routing, tool execution, and controlled interaction between the LLM and external tools.
- Containerized the application using **Docker** and developed a **GitHub Actions CI/CD pipeline** to automate code validation, linting, Docker image builds, and publishing to Docker Hub.
- Deployed the containerized application to **AWS EC2** using a self-hosted GitHub Actions runner, with deployment health checks, container logging, and best-effort rollback.
- Managed API credentials and application configuration through environment variables and GitHub Actions secrets.

Tech Stack: Python, ReactJS, LangGraph, LangChain, LangSmith, RAG, HITL, LLMs, Docker, Docker Hub, GitHub Actions, AWS EC2, Linux, REST APIs, Ruff, pytest.

Microsoft Azure AI & Generative AI – Hands-on Projects
Jan 2026 – April 2026

- Azure AI Foundry & Azure OpenAI Integration: Configured Azure AI Foundry resources and model deployments, integrating Azure-hosted language models with Python applications using the OpenAI SDK and environment-based configuration.
- Text-to-Speech (TTS) and Speech-to-Text (STT): Implemented audio generation using gpt-4o-mini-tts and explored speech transcription using the Whisper deployment for audio-to-text processing.
- LLM Function Calling & Tool Integration: Developed a Python-based weather assistant using Azure-hosted LLMs and function calling to identify user intent, invoke external APIs, process tool responses, and generate natural-language answers.
- External API Integration: Integrated the OpenWeatherMap API to retrieve current weather information, including temperature, humidity, weather conditions, and wind speed.
- Python Application Configuration: Used Python, python-dotenv, environment variables, and API-based integrations to manage application configuration and credentials.

Tech Stack: Microsoft Azure AI Foundry, Azure OpenAI, Python, OpenAI SDK, Function Calling, REST APIs, Whisper, Text-to-Speech, Speech-to-Text, OpenWeatherMap API.

Real-Time LLM Fine-Tuning & GPU Inference Platform Nov 2025 - Dec 2025

- Fine-tuned **TinyLlama-1.1B-Chat-v1.0** model on a domain-specific HR Question & Answer dataset using **Parameter-Efficient Fine-Tuning (PEFT)** techniques to improve task-specific response quality.
- Configured and managed model training and inference on a **HyperStack VM** powered by **NVIDIA H100 GPU**, enabling high-performance LLM experimentation and deployment.
- Developed a **Dockerized FastAPI** inference service to expose the fine-tuned model through scalable REST APIs for real-time AI applications.
- Established a secure remote development workflow using **SSH-based authentication** and deployment tools for efficient GPU resource management.

Tech Stack: Python, TinyLlama-1.1B-Chat, FastAPI, Docker, Hyperstack VM, NVIDIA H100 GPU, SSH Key, WinSCP, PEFT.

Trip Mate-AI Multi-Agent Travel Planner Aug 2025 - Nov 2025

- Designed and developed an agentic AI travel-planning application using LangGraph to orchestrate specialized agents for flight search, hotel search, itinerary generation, and final response synthesis.
- Built a FastAPI backend with REST endpoints, Pydantic request validation, and a web-based interface for submitting travel requests and displaying AI-generated responses.
- Implemented shared graph state management and PostgreSQL-backed checkpointing to support persistent workflow state and conversation continuity.
- Integrated Groq-hosted LLM inference for natural-language processing, travel-planning reasoning, and consolidated itinerary generation.
- Containerized the application using Docker, with environment-based configuration and .dockerignore rules for reproducible deployment and improved secret handling.

Tech Stack: Python, LangGraph, LangChain, Groq LLM, FastAPI, PostgreSQL, HTML, CSS, JavaScript, Docker

Automatic Target Detection (ATD) & Recognition (ATR) System Jan 2024 - Jul 2025

- Developed an end-to-end AI-powered Automatic Target Detection & Recognition (ATD/ATR) system for defense applications, enabling detection of tanks, aircraft, ships, and military vehicles from EO, IR, and SAR imagery.
- Fine-tuned **REDET** and **STD+HViT** models on **DOTA** and custom datasets using multiple backbones (ResNet, ViT, HViT, Oriented Faster R-CNN) to imporve object detection accuracy and robustness.
- Designed a complete data pipeline covering dataset acquisition, annotation, preprocessing (tiling, denoising, contrast enhancement), model training, evaluation, and inference in an offline environment.
- Built a containerized deployment workflow using Docker on NVIDIA DGX A100, ensuring scalable and reproducible model training and inference for defense environments.

Tech Stack: Python, PyTorch, REDET, STD+HViT, Oriented Faster R-CNN, Mask R-CNN, DOTA, Roboflow, Google Earth, UMBRA (SAR), RoLabeling, X-AnyLabeling, Docker, NVIDIA DGX A100, Linux

EDUCATION

Master of Computer Applications (MCA) Aug 2016 - Apr 2019

St. Joseph's College (Autonomous), Tiruchirappalli

Bachelor of Science - Physics (BSc) Aug 2013 - Apr 2016

St. Joseph's College (Autonomous), Tiruchirappalli